Topic: Nonpolynomial coefficients

Question: Find the first three terms of the power series solutions of the differential equation around the ordinary point $x_0 = 0$.

$$y'' - (\cos x)y = 0$$

Answer choices:

A $$y = c_0\left(1 + \frac{1}{2}x^2 - \frac{1}{144}x^6 + \ldots\right) + c_1\left(x + \frac{1}{6}x^3 - \frac{1}{60}x^5 + \ldots\right)$$

B $$y = c_0\left(1 + \frac{1}{6}x^2 + \frac{1}{24}x^6 + \ldots\right) + c_1\left(x + \frac{1}{2}x^3 + \frac{1}{6}x^5 + \ldots\right)$$

C $$y = c_0\left(1 + \frac{1}{2}x^2 + \ldots\right) + c_1\left(x + \frac{1}{6}x^3 + \ldots\right)$$

D $$y = c_0\left(1 + \frac{1}{6}x^2 + \ldots\right) + c_1\left(x + \frac{1}{2}x^3 + \ldots\right)$$

© Krista King Math

Solution: A

We know $x_0 = 0$ is an ordinary point, because $\cos x$ is analytic there. And we know that the Maclaurin series representation of $\cos x$ is

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots$$

So we can substitute into the differential equation to get

$$y'' - (\cos x)y = 0$$

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} - \left(1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots\right) \sum_{n=0}^{\infty} c_n x^n = 0$$

Now we'll just expand each series through its first few terms,

$$(2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + 30c_6 x^4 + \dots)$$

$$-\left(1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots\right)(c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots) = 0$$

$$(2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + 30c_6 x^4 + \dots)$$

$$-(c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots)$$

$$+\frac{1}{2!}x^2(c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots)$$

$$-\frac{1}{4!}x^4(c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots)$$

$$+\frac{1}{6!}x^6(c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots) + \dots = 0$$

© Krista King Math

$$(2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + 30c_6x^4 + \dots)$$

$$-(c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \dots)$$

$$+\left(\frac{1}{2!}c_0x^2 + \frac{1}{2!}c_1x^3 + \frac{1}{2!}c_2x^4 + \frac{1}{2!}c_3x^5 + \frac{1}{2!}c_4x^6 + \dots\right)$$

$$-\left(\frac{1}{4!}c_0x^4 + \frac{1}{4!}c_1x^5 + \frac{1}{4!}c_2x^6 + \frac{1}{4!}c_3x^7 + \frac{1}{4!}c_4x^8 + \dots\right)$$

$$+\left(\frac{1}{6!}c_0x^6 + \frac{1}{6!}c_1x^7 + \frac{1}{6!}c_2x^8 + \frac{1}{6!}c_3x^9 + \frac{1}{6!}c_4x^{10} + \dots\right) + \dots = 0$$

and then collect equivalent powers of x.

$$(2c_2 - c_0) + (6c_3 - c_1)x + \left(12c_4 - c_2 + \frac{1}{2!}c_0\right)x^2$$

$$+\left(20c_5 - c_3 + \frac{1}{2!}c_1\right)x^3 + \left(30c_6 - c_4 + \frac{1}{2!}c_2 - \frac{1}{4!}c_0\right)x^4 + \dots$$

From this result, we get a system of equations.

$$2c_2 - c_0 = 0$$

$$6c_3 - c_1 = 0$$

$$12c_4 - c_2 + \frac{1}{2!}c_0 = 0$$

$$20c_5 - c_3 + \frac{1}{2!}c_1 = 0$$

© Krista King Math



$$30c_6 - c_4 + \frac{1}{2!}c_2 - \frac{1}{4!}c_0 = 0$$

which simplifies to

$$c_2 = \frac{1}{2}c_0 \qquad c_3 = \frac{1}{6}c_1$$

$$c_4 = 0 \qquad c_5 = -\frac{1}{60}c_1$$

$$c_6 = -\frac{1}{144}c_0$$

Pull these coefficients together into the power series solution.

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + c_7x^7 + \dots$$

$$y = c_0 + c_1x + \frac{1}{2}c_0x^2 + \frac{1}{6}c_1x^3 + 0x^4 - \frac{1}{60}c_1x^5 + \dots$$

$$y = c_0\left(1 + \frac{1}{2}x^2 - \frac{1}{144}x^6 + \dots\right) + c_1\left(x + \frac{1}{6}x^3 - \frac{1}{60}x^5 + \dots\right)$$

© Krista King Math

Topic: Nonpolynomial coefficients

Question: Find the first four terms of the power series solutions of the differential equation around the ordinary point $x_0 = 0$.

$$y'' + e^x y' - y = 0$$

Answer choices:

A $$y = c_0\left(x - \frac{1}{2}x^2 + \frac{1}{6}x^3 - \frac{1}{24}x^4 + \ldots\right) + c_1\left(1 + \frac{1}{2}x^2 - \frac{1}{6}x^3 - \frac{1}{120}x^5 + \ldots\right)$$

B $$y = c_0\left(1 + \frac{1}{2}x^2 - \frac{1}{6}x^3 - \frac{1}{120}x^5 + \ldots\right) + c_1\left(x - \frac{1}{2}x^2 + \frac{1}{6}x^3 - \frac{1}{24}x^4 + \ldots\right)$$

C $$y = c_0\left(x - \frac{1}{2}x^2 + \frac{1}{6}x^3 - \frac{1}{24}x^4 + \ldots\right) - c_1\left(1 - \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{120}x^5 - \ldots\right)$$

D $$y = c_0\left(1 - \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{120}x^5 - \ldots\right) - c_1\left(x - \frac{1}{2}x^2 + \frac{1}{6}x^3 - \frac{1}{24}x^4 + \ldots\right)$$

© Krista King Math

Solution: B

We know $x_0 = 0$ is an ordinary point, because e^x is analytic there. And we know that the Maclaurin series representation of e^x is

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \dots$$

So we can substitute into the differential equation to get

$$y'' + e^x y' - y = 0$$

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \left(1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \dots\right) \sum_{n=1}^{\infty} c_n n x^{n-1}$$

$$-\sum_{n=0}^{\infty} c_n x^n = 0$$

Now we'll just expand each series through its first few terms,

$$(2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + 30c_6 x^4 + \dots)$$

$$+\left(1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \dots\right)(c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + 5c_5 x^4 + \dots)$$

$$-(c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots) = 0$$

$$(2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + 30c_6 x^4 + \dots)$$

$$+(c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + 5c_5 x^4 + \dots)$$

$$+x(c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + 5c_5 x^4 + \dots)$$

© Krista King Math

$$+\frac{1}{2}x^2(c_1 + 2c_2x + 3c_3x^2 + 4c_4x^3 + 5c_5x^4 + \dots)$$

$$+\frac{1}{6}x^3(c_1 + 2c_2x + 3c_3x^2 + 4c_4x^3 + 5c_5x^4 + \dots)$$

$$+\frac{1}{24}x^4(c_1 + 2c_2x + 3c_3x^2 + 4c_4x^3 + 5c_5x^4 + \dots)$$

$$-(c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \dots) = 0$$

$$(2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + 30c_6x^4 + \dots)$$

$$+(c_1 + 2c_2x + 3c_3x^2 + 4c_4x^3 + 5c_5x^4 + \dots)$$

$$+(c_1x + 2c_2x^2 + 3c_3x^3 + 4c_4x^4 + 5c_5x^5 + \dots)$$

$$+\left(\frac{1}{2}c_1x^2 + c_2x^3 + \frac{3}{2}c_3x^4 + 2c_4x^5 + \frac{5}{2}c_5x^6 + \dots\right)$$

$$+\left(\frac{1}{6}c_1x^3 + \frac{1}{3}c_2x^4 + \frac{1}{2}c_3x^5 + \frac{2}{3}c_4x^6 + \frac{5}{6}c_5x^7 + \dots\right)$$

$$+\left(\frac{1}{24}c_1x^4 + \frac{1}{12}c_2x^5 + \frac{1}{8}c_3x^6 + \frac{1}{6}c_4x^7 + \frac{5}{24}c_5x^8 + \dots\right)$$

$$-(c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \dots) = 0$$

and then collect equivalent powers of x.

$$(2c_2 + c_1 - c_0) + (6c_3 + 2c_2)x + \left(12c_4 + 3c_3 + c_2 + \frac{1}{2}c_1\right)x^2$$

© Krista King Math

$$+\left(20c_5+4c_4+2c_3+c_2+\frac{1}{6}c_1\right)x^3$$

$$+\left(30c_6+5c_5+3c_4+\frac{3}{2}c_3+\frac{1}{3}c_2+\frac{1}{24}c_1\right)x^4+\ldots$$

From this result, we get a system of equations.

$$2c_2+c_1-c_0=0$$

$$6c_3+2c_2=0$$

$$12c_4+3c_3+c_2+\frac{1}{2}c_1=0$$

$$20c_5+4c_4+2c_3+c_2+\frac{1}{6}c_1=0$$

$$30c_6+5c_5+3c_4+\frac{3}{2}c_3+\frac{1}{3}c_2+\frac{1}{24}c_1=0$$

which simplifies to

$$c_2=-\frac{1}{2}c_1+\frac{1}{2}c_0 \qquad c_3=\frac{1}{6}c_1-\frac{1}{6}c_0$$

$$c_4=-\frac{1}{24}c_1 \qquad c_5=\frac{1}{120}c_1-\frac{1}{120}c_0$$

$$c_6=-\frac{1}{720}c_1+\frac{1}{240}c_0$$

Pull these coefficients together into the power series solution.

$$y=c_0+c_1x+c_2x^2+c_3x^3+c_4x^4+c_5x^5+c_6x^6+c_7x^7+\ldots$$

© Krista King Math

$$y = c_0 + c_1 x + \left(-\frac{1}{2}c_1 + \frac{1}{2}c_0\right)x^2 + \left(\frac{1}{6}c_1 - \frac{1}{6}c_0\right)x^3 + \left(-\frac{1}{24}c_1\right)x^4$$

$$+\left(\frac{1}{120}c_1 - \frac{1}{120}c_0\right)x^5 + \left(-\frac{1}{720}c_1 + \frac{1}{240}c_0\right)x^6 + \ldots$$

$$y = c_0\left(1 + \frac{1}{2}x^2 - \frac{1}{6}x^3 - \frac{1}{120}x^5 + \frac{1}{240}x^6 - \ldots\right)$$

$$+c_1\left(x - \frac{1}{2}x^2 + \frac{1}{6}x^3 - \frac{1}{24}x^4 + \frac{1}{120}x^5 - \frac{1}{720}x^6 + \ldots\right)$$

Simplify to just the first four terms.

$$y = c_0\left(1 + \frac{1}{2}x^2 - \frac{1}{6}x^3 - \frac{1}{120}x^5 + \ldots\right) + c_1\left(x - \frac{1}{2}x^2 + \frac{1}{6}x^3 - \frac{1}{24}x^4 + \ldots\right)$$

© Krista King Math

Topic: Nonpolynomial coefficients

Question: Find the first three terms of the power series solutions of the differential equation around the ordinary point $x_0 = 0$.

$$y'' - y' + \ln(1+x)y = 0$$

Answer choices:

A $$y = c_0\left(x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \ldots\right) + c_1\left(1 - \frac{1}{6}x^3 - \frac{1}{60}x^5 + \ldots\right)$$

B $$y = c_0\left(1 - \frac{1}{6}x^3 - \frac{1}{60}x^5 + \ldots\right) + c_1\left(x + \frac{1}{2}x^2 + \frac{1}{6}x^3 - \ldots\right)$$

C $$y = c_0\left(x - \frac{1}{2}x^2 - \frac{1}{6}x^3 - \ldots\right) + c_1\left(1 + \frac{1}{6}x^3 + \frac{1}{60}x^5 + \ldots\right)$$

D $$y = c_0\left(1 + \frac{1}{6}x^3 + \frac{1}{60}x^5 + \ldots\right) + c_1\left(x - \frac{1}{2}x^2 - \frac{1}{6}x^3 - \ldots\right)$$

© Krista King Math

Solution: B

We know $x_0 = 0$ is an ordinary point, because $\ln(1+x)$ is analytic there. And we know that the Maclaurin series representation of $\ln(1+x)$ is

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5 - \dots$$

So we can substitute into the differential equation to get

$$y'' - y' + \ln(1+x)y = 0$$

$$\sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} - \sum_{n=1}^{\infty} c_n n x^{n-1} + \ln(1+x) \sum_{n=0}^{\infty} c_n x^n = 0$$

Now we'll just expand each series through its first few terms,

$$(2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + 30c_6x^4 + \dots)$$

$$-(c_1 + 2c_2x + 3c_3x^2 + 4c_4x^3 + 5c_5x^4 + \dots)$$

$$+\left(x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5 - \dots\right)(c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \dots) = 0$$

$$(2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + 30c_6x^4 + \dots)$$

$$-(c_1 + 2c_2x + 3c_3x^2 + 4c_4x^3 + 5c_5x^4 + \dots)$$

$$+x(c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \dots)$$

$$-\frac{1}{2}x^2(c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \dots)$$

© Krista King Math

$$+\frac{1}{3}x^3(c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \dots)$$

$$-\frac{1}{4}x^4(c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \dots)$$

$$+\frac{1}{5}x^5(c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \dots) - \dots = 0$$

$$2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + 30c_6x^4 + \dots$$

$$-c_1 - 2c_2x - 3c_3x^2 - 4c_4x^3 - 5c_5x^4 - \dots$$

$$+c_0x + c_1x^2 + c_2x^3 + c_3x^4 + c_4x^5 + \dots$$

$$-\frac{1}{2}c_0x^2 - \frac{1}{2}c_1x^3 - \frac{1}{2}c_2x^4 - \frac{1}{2}c_3x^5 - \frac{1}{2}c_4x^6 - \dots$$

$$+\frac{1}{3}c_0x^3 + \frac{1}{3}c_1x^4 + \frac{1}{3}c_2x^5 + \frac{1}{3}c_3x^6 + \frac{1}{3}c_4x^7 + \dots$$

$$-\frac{1}{4}c_0x^4 - \frac{1}{4}c_1x^5 - \frac{1}{4}c_2x^6 - \frac{1}{4}c_3x^7 - \frac{1}{4}c_4x^8 - \dots$$

$$+\frac{1}{5}c_0x^5 + \frac{1}{5}c_1x^6 + \frac{1}{5}c_2x^7 + \frac{1}{5}c_3x^8 + \frac{1}{5}c_4x^9 + \dots = 0$$

and then collect equivalent powers of x.

$$(2c_2 - c_1) + (6c_3 - 2c_2 + c_0)x + \left(12c_4 - 3c_3 + c_1 - \frac{1}{2}c_0\right)x^2$$

$$+\left(20c_5 - 4c_4 + c_2 - \frac{1}{2}c_1 + \frac{1}{3}c_0\right)x^3$$

© Krista King Math

$$+\left(30c_6 - 5c_5 + c_3 - \frac{1}{2}c_2 + \frac{1}{3}c_1 - \frac{1}{4}c_0\right)x^4 + \ldots = 0$$

From this result, we get a system of equations.

$$2c_2 - c_1 = 0$$

$$6c_3 - 2c_2 + c_0 = 0$$

$$12c_4 - 3c_3 + c_1 - \frac{1}{2}c_0 = 0$$

$$20c_5 - 4c_4 + c_2 - \frac{1}{2}c_1 + \frac{1}{3}c_0 = 0$$

$$30c_6 - 5c_5 + c_3 - \frac{1}{2}c_2 + \frac{1}{3}c_1 - \frac{1}{4}c_0 = 0$$

which simplifies to

$$c_2 = \frac{1}{2}c_1 \qquad c_3 = \frac{1}{6}c_1 - \frac{1}{6}c_0$$

$$c_4 = -\frac{1}{24}c_1 \qquad c_5 = -\frac{1}{120}c_1 - \frac{1}{60}c_0$$

$$c_6 = -\frac{7}{720}c_1 + \frac{1}{90}c_0$$

Pull these coefficients together into the power series solution.

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + c_7x^7 + \ldots$$

$$y = c_0 + c_1x + \frac{1}{2}c_1x^2 + \left(\frac{1}{6}c_1 - \frac{1}{6}c_0\right)x^3 - \frac{1}{24}c_1x^4$$

© Krista King Math

$$+\left(-\frac{1}{120}c_1-\frac{1}{60}c_0\right)x^5+\left(-\frac{7}{720}c_1+\frac{1}{90}c_0\right)x^6+\ldots$$

$$y=c_0\left(1-\frac{1}{6}x^3-\frac{1}{60}x^5+\frac{1}{90}x^6+\ldots\right)$$

$$+c_1\left(x+\frac{1}{2}x^2+\frac{1}{6}x^3-\frac{1}{24}x^4-\frac{1}{120}x^5-\frac{7}{720}x^6+\ldots\right)$$

Simplify to just the first three terms.

$$y=c_0\left(1-\frac{1}{6}x^3-\frac{1}{60}x^5+\ldots\right)+c_1\left(x+\frac{1}{2}x^2+\frac{1}{6}x^3-\ldots\right)$$

© Krista King Math